EDS223 - Homework Assignment 1

Air quality as measured by particulate matter 2.5ppm across Santa Cruz and San Mateo counties, in northern California

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Air quality as measured by particulate matter 2.5ppm across Santa Cruz and San Mateo counties.

The two maps presents demonstrate different levels of PM2.5 in San Mateo County to the north, and Santa Cruz County to the south. The two counties were choosen due to prior knowledge of the region and counties, as well as interest in how the two counties vary in terms of PM2.5 levels. PM2.5, which often is the result of industrial activities, as well as vehicular activities among other things, was choosen as a attribute due to both concern over air quality in the region, as well as a significant amount of variance in the types of environment that can be found across the region; much of the middle section of the regions (along the border of San Mateo and Santa Cruz) are fairly rural, while the northern portion of the region is heavily influenced by urbanisation and the southern portion of the region is primarily farmland. Cumulatively, the levels of PM2.5 are higher across San Mateo County when compared to Santa Cruz County. Northern San Mateo County borders San Francisco and South San Francisco, while south-eastern San Mateo borders Santa Clara County. To the west is the Pacific Ocean; this is of note considering that some of the highest levels of PM2.5 within the county are along the western border. San Mateo County has a contrasting mix of both urban areas, as well

as a busy highway that serves as a conduit for travel to/from San Francisco, and large open areas that are a mix of small farms and state park land. Northern Santa Cruz County borders southern San Mateo County, and consists primarily of open land, state park land, and small farms. As with San Mateo County, the Pacific Ocean sits to the west of the county; the PM2.5 levels are missing for this block of the county, while Santa Clara County sits to the east of Santa Cruz County, and Monterey County is to the south. Away from the coast, most of the county is fairly forest and pastoral, with a large farming area to the south along the Monterey County border. The highest levels of PM2.5 are along part of the border with Santa Clara County, near the town of Los Gatos.

1 Load the libraries

```
library(here)
```

```
here() starts at /home/eowen/MEDS/eds223/EDS223-HW1
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(sf)
```

```
Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(stars)
```

```
Loading required package: abind
```

```
library(tmap)
```

2 Load the data

```
ej_screen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_C
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/eowen/MEDS/eds223/EDS223-HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

3 Filter the data to California

3.1 Filter the data to Santa Cruz County, San Mateo County

```
california <- ej_screen %>%
  dplyr::filter(ST_ABBREV == "CA")

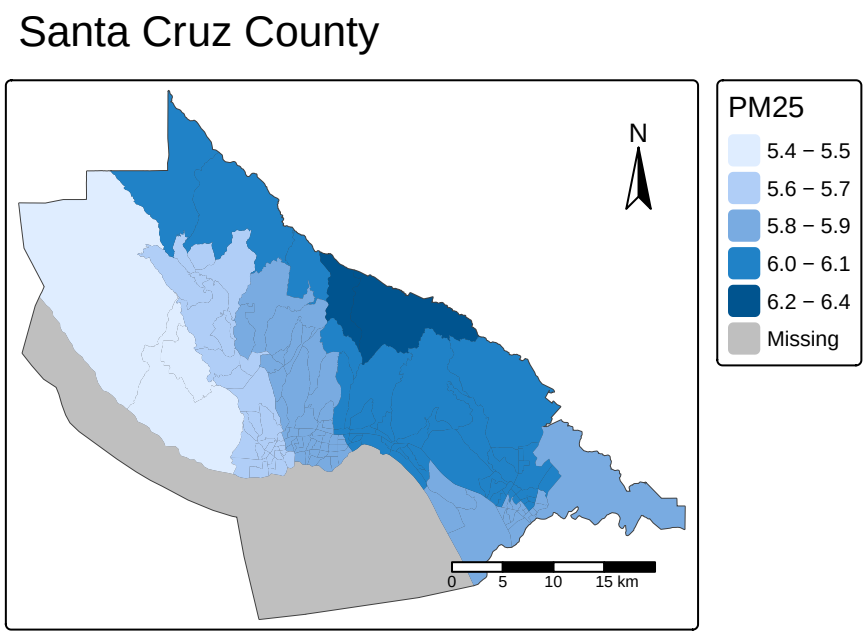
santa_cruz<- california %>%
  dplyr::filter(CNTY_NAME %in% c("Santa Cruz County"))

san_mateo <- california %>%
    dplyr::filter(CNTY_NAME %in% c("San Mateo County"))
```

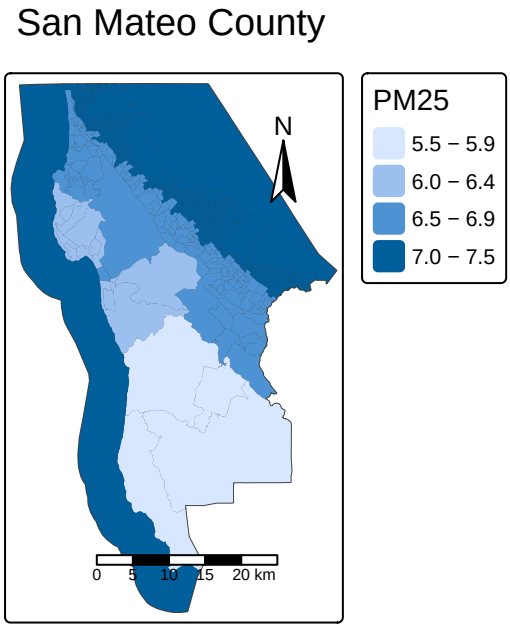
4 Build maps and label maps with PM2.5

```
tm_shape(santa_cruz)+
    tm_borders()+
    tm_fill("PM25")+
    tm_title("Santa Cruz County")+
```

```
tm_compass(position = c('right', 'top'))+
tm_scalebar()
```




```
tm_shape(san_mateo)+
    tm_borders()+
    tm_fill("PM25") +
    tm_title("San Mateo County")+
    tm_compass(position = c('right', 'top'))+
    tm_scalebar()
```



5 Revision Memo

The specification that was missed was 'Brief Explanation Needs More Explanation'. Within the introductory paragraph, an explanation on why the counties of San Mateo and Santa Cruz has been added, as well as why PM2.5 was choosen as the attribute under revision.